

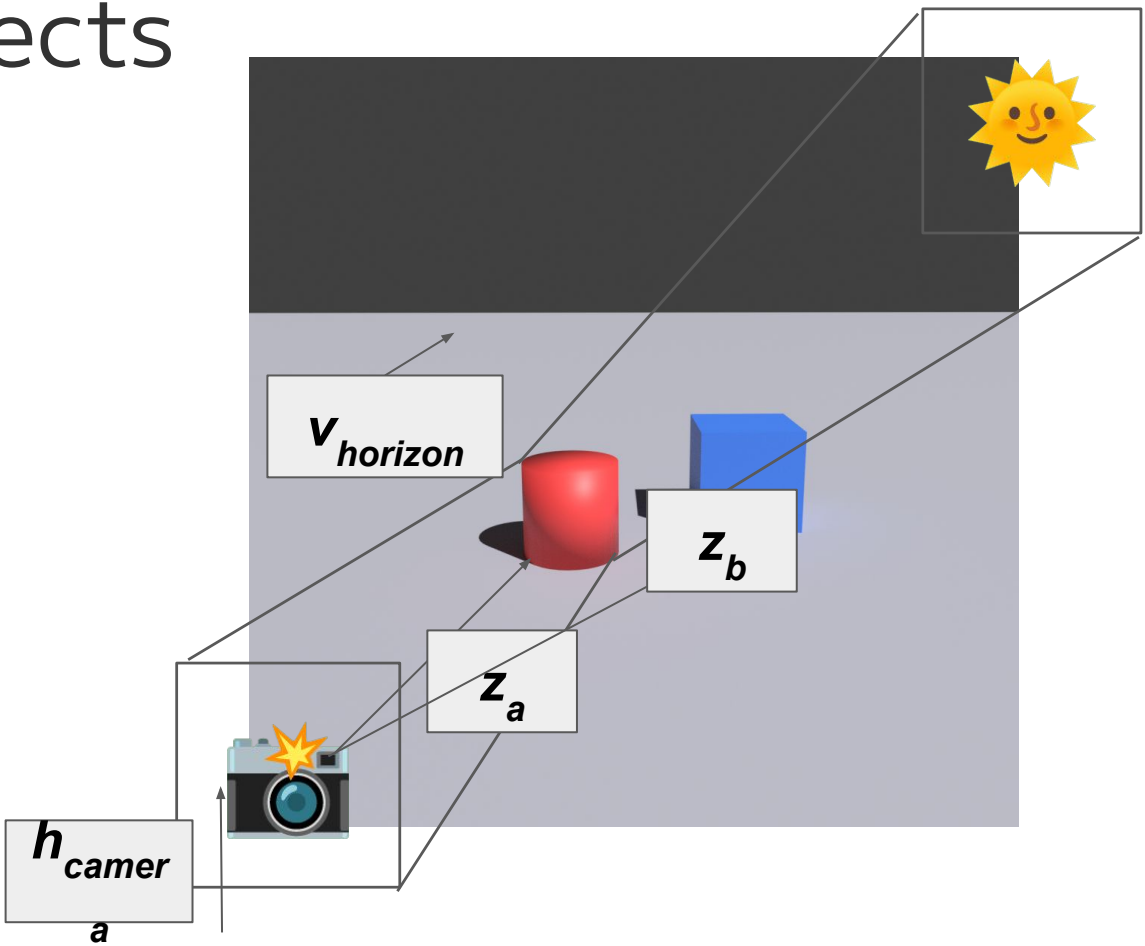
Motivation

- **Metric depth, distance in metres, is key to 3D understanding.**
 - For robotics and 3D reconstruction [Atar+, IROS'25; Yin+, ICCV'23].
 - **Do VLMs internally represent metric depth and use it?**
 - We probe if depth is decodable beyond simple visual cues.
 - We intervene to test if it guides answers.
- 😊 Spatial relations are decodable and causally effective in VLMs. (cf. 2D relation [Kang+, ICLR'26], occlusion [Maheshwari+, arXiv'26], 3D topology [Wang & Gao, arXiv'26])
- 😞 Yet metric depth remains underexplored inside VLMs.
- 😞 Depth is strongly correlated with 2D cues. (e.g., Nearer objects appear lower and larger.)
- ➔ Prior work breaks that tie at render time [Min+, ECCV'26]; whereas we benchmark against these cues.

Key Findings

- Metric depth is decodable at nearly every layer, beyond what image height and size alone can provide.
- But its causal effect is confined to an early LM layer range.

Task Definition

- We give the VLM an image with two objects at different depths z_A and z_B .
 - We ask
 - **Q:** "Which one's centre is a shorter distance from the camera?"
 - **A:** A or B
 - and record its answer.
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Dataset Construction

- We generated 1,620 synthetic scene.
 - two objects on the ground plane.
 - known depth z , under controlled camera and lighting.
- On a ground plane, depth z and image position v are tied, and the relationship can be written as:

$$\frac{v - v_{\text{horizon}}}{\text{how far below the horizon (px)}} \propto \frac{f \cdot h_{\text{camera}}}{z} \quad \begin{matrix} f & : & \text{focal length (px)} \\ h_{\text{camera}} & : & \text{camera height (m)} \end{matrix}$$

z : metric depth (m)

Conditions

- To reduce bias, we generated data under 3 conditions (540 each)
 - same apparent size, depth differs
 - each object appears both near and far position
 - random sampling, no additional constraints

Construction Process

1. Sample a camera pose, height, pitch and yaw jittered.
2. Sample the pair's depth at a depth ratio. (depth range 2.95–6.65 m)
3. Render and record ground truth information. (depth z , mask and horizontal row)

Results

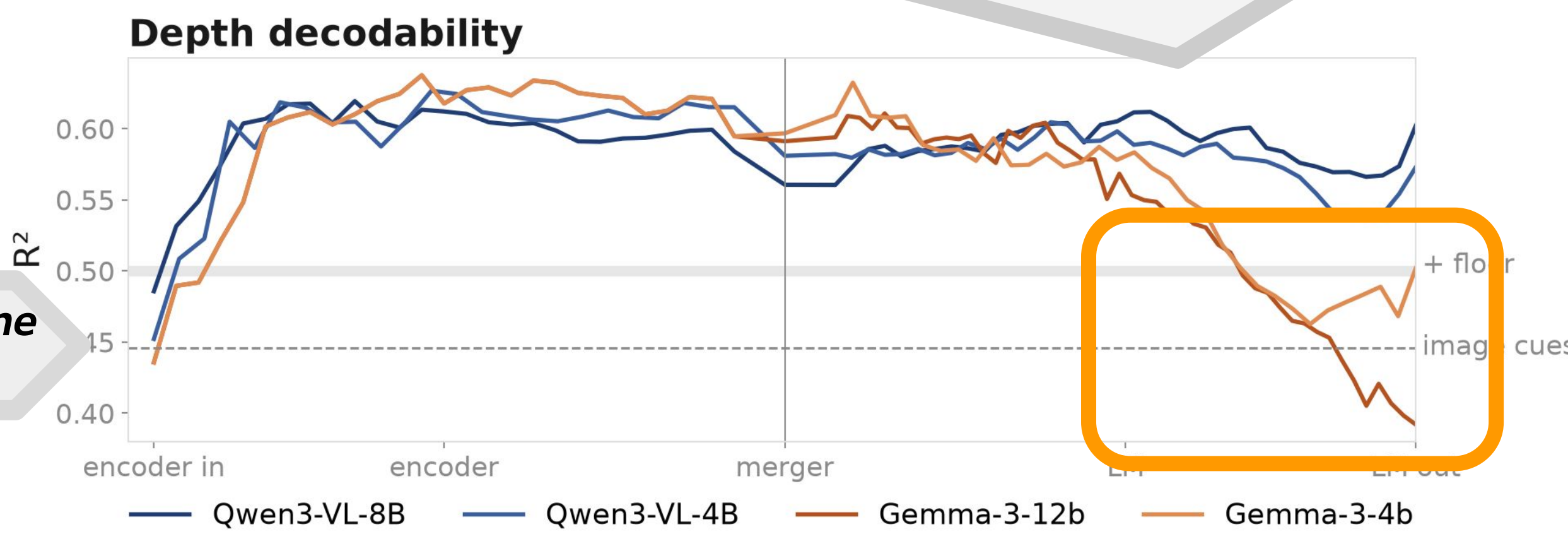
Synthetic data accuracy

model	acc.
Qwen3-VL-8B [Bai+, arXiv'25]	0.947
Qwen3-VL-4B [Bai+, arXiv'25]	0.863
Gemma-3-12b [Gemma Team, arXiv'25]	0.600
Gemma-3-4b [Gemma Team, arXiv'25]	0.515

lower task accuracy lose decodable metric depth in later LM layers.

Probing

Most layers successfully decoded metric depth



Analysis Protocol

Probing

- To analyze if the model represents depth via linear probing:
 1. We pool the object's visual-token activations at each layer.
 2. We fit a linear model to predict its depth in metres.
- ➔ High $R^2 \Rightarrow$ depth is decodable

$$\hat{z} = w^T h + b$$

$\hat{z}$: predicted metric depth
 w : direction used to decode depth
 h : pooled object's visual-token activations
 b : bias



- Probe performance may reflect simple visual cues, not depth.
 - ➔ For comparison, we fit a linear model using five visual cues:
 - ① height below horizon
 - ② horizontal off-centre
 - ③ apparent height
 - ④ apparent area
 - ⑤ bbox fill

Intervention

- To analyze if internal depth representations guide answers:
 1. We shift decoded depth using the probe model.
 - Probe direction defines the depth axis.
 - We shift the object state along it by α metres.

$$h_t \leftarrow h_t + \alpha \frac{w}{w^T w}, t \in \mathcal{T}_{\text{obj}}$$

$\mathcal{T}_{\text{obj}} = \{t_1, t_2, \dots, t_{|\mathcal{T}_{\text{obj}}|}\}$: set of the object's visual-token indices
 h_t : activation vector at visual token t

2. We read the A/B logits, how strongly the model favors each, and define the **margin** as their difference.

$$\text{margin} = \ell_{\text{target}}^{(\alpha)} - \ell_{\text{other}}^{(\alpha)}$$

$\ell_{\text{target}}^{(\alpha)}$: target-answer logit after an α -metre depth shift
 $\ell_{\text{other}}^{(\alpha)}$: other-answer logit after an α -metre depth shift

3. We measure belief gap Δ between farther and nearer shifts.

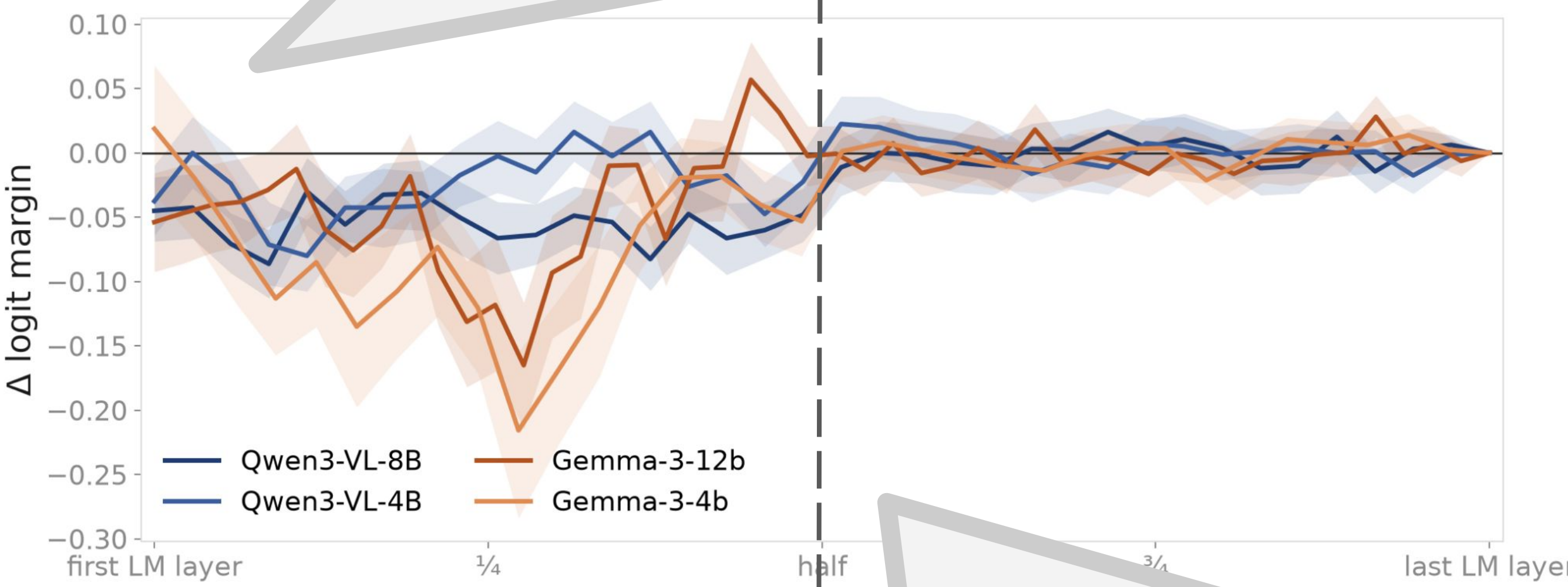
$$\Delta = \text{margin}_{+\alpha} - \text{margin}_{-\alpha}$$

pushed farther pushed nearer

➔ High belief gap $\Delta \Rightarrow$ The model uses internal depth

Intervention

Negative Δ is preferred because we are asking for shorter distance



Intervention in an early LM layer shifts the model's belief in the expected direction

Discussion & Future Work

- Early-LM Intervention selectively shifts belief, but does not show that depth is used for the final decision.
 - ➔ The intervention effect is too small to cross the decision margin and also appears without depth-dependent behavior.
- Metric depth cannot be fully separated from its 2D cues in the current dataset.
- The relationship between objects is not considered.